Speech Emotion Recognition

RAVDESS Dataset · Multi-Branch CNN · MFCC + Mel + Chroma

Deep Learning Project | TensorFlow / Keras

Problem Overview

Goal

Automatically classify human speech audio into 7 emotion categories.

Dataset

RAVDESS: 24 professional actors, 7 emotions × ~104 samples each.

Input

4-second audio clips at 22,050 Hz sample rate.

Labels

Neutral · Happy · Sad · Angry · Fearful · Disgust · Surprised

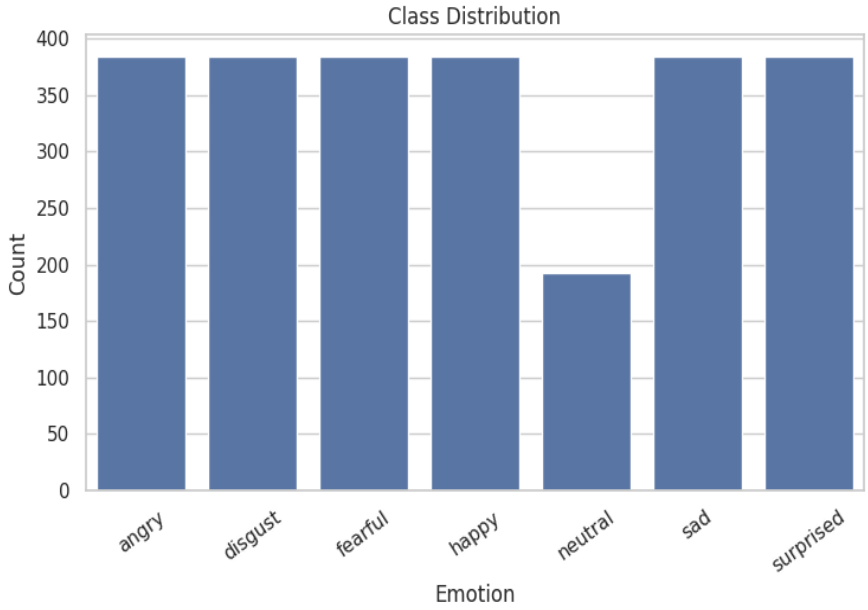
Dataset & Exploratory Analysis

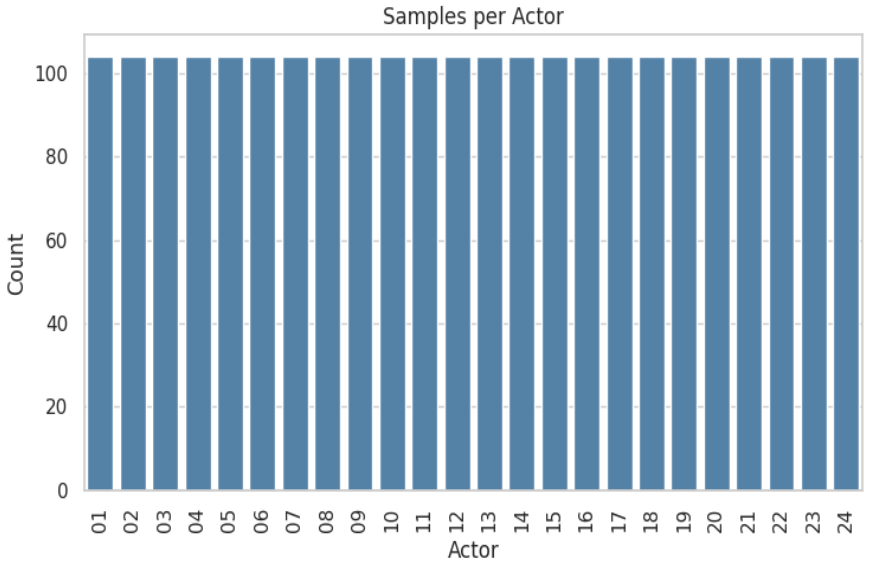
1,248
Total Samples

24
Actors

7
Emotions

4 sec
Clip Duration





Audio Preprocessing Pipeline

01 **Load Audio**

22,050 Hz, up to 4 seconds

02 **Fix Length**

Pad with zeros or truncate to exactly 88,200 samples

03 **Feature Extract**

MFCC (40) · Mel Spectrogram (128) · Chroma (12)

04 **Pad Sequences**

Align all time-steps to MAX_LEN=300 frames

05 **Standardize**

StandardScaler fitted on train frames only

Audio Feature Extraction

MFCC

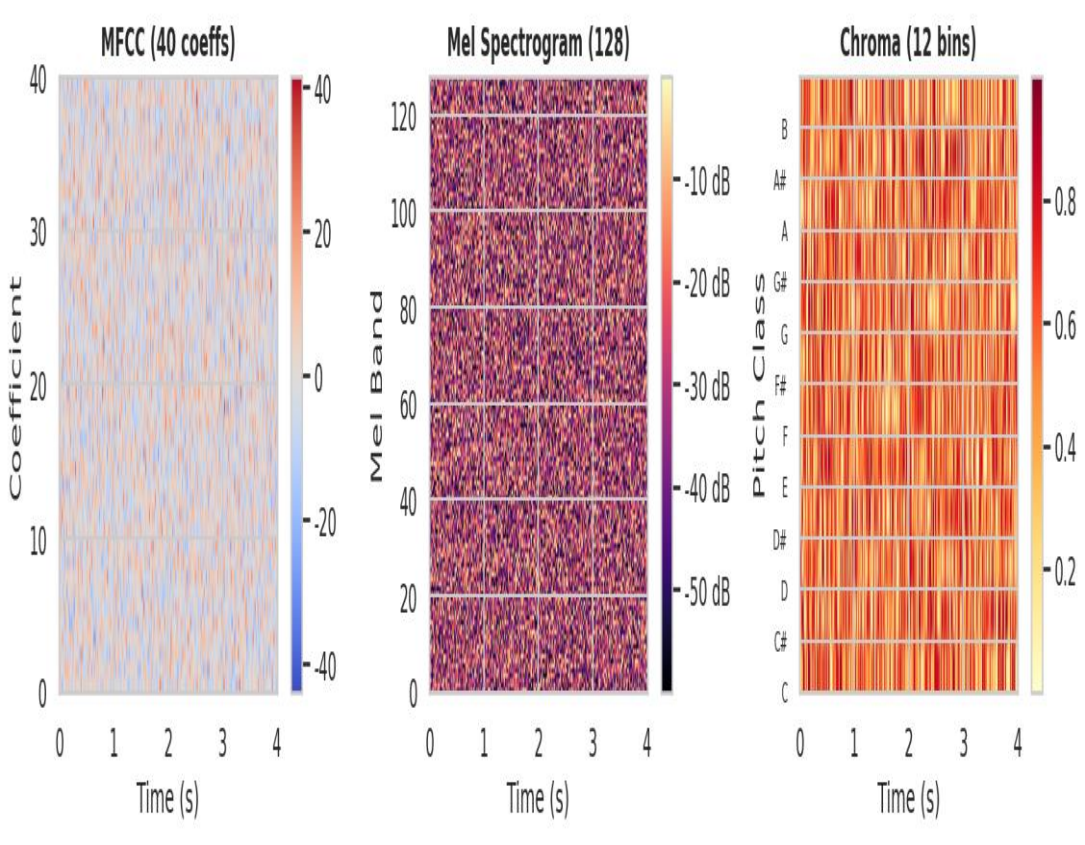
40 cepstral coefficients capture the overall shape of the spectral envelope — encoding timbre and vocal tract characteristics.

Mel Spectrogram

128-band frequency representation on a perceptually-scaled axis — captures energy distribution across emotional speech.

Chroma

12 pitch-class features representing the harmonic content and tonal qualities associated with different emotions.



Data Augmentation Strategy

Why Augment?

With ~1,248 samples, overfitting is a real risk. Augmentation synthetically triples the training set by applying transforms that preserve emotional content.

Add Gaussian Noise

Params: factor = 0.005

Simulates microphone noise and real-world recording conditions.

Time Shift

Params: shift ≤ ±0.25 s

Shifts the audio forward/backward to reduce temporal position bias.

✓ Augmentation applied to training set only · Methods used: Noise + Time Shift · Training tripled from ~1996 → ~5988 samples

Multi-Branch CNN Architecture

MFCC Branch

Input (300, 40)

Conv1D(128, k=5) + BN + ReLU + Pool + Drop(0.4)

Conv1D(64, k=3) + BN + ReLU + Pool + Drop(0.3)

Conv1D(32, k=3) + BN + ReLU + Pool + Drop(0.2)

GlobalAvgPooling1D

Mel Branch

Input (300, 128)

Conv1D(128, k=5) + BN + ReLU + Pool + Drop(0.4)

Conv1D(64, k=3) + BN + ReLU + Pool + Drop(0.3)

Conv1D(32, k=3) + BN + ReLU + Pool + Drop(0.2)

GlobalAvgPooling1D

Chroma Branch

Input (300, 12)

Conv1D(128, k=5) + BN + ReLU + Pool + Drop(0.4)

Conv1D(64, k=3) + BN + ReLU + Pool + Drop(0.3)

Conv1D(32, k=3) + BN + ReLU + Pool + Drop(0.2)

GlobalAvgPooling1D

Concatenate → Dense(128, ReLU) → Dropout(0.4)

Softmax Output (7 classes)

Training Configuration & Callbacks

Hyperparameters

Optimizer	Adam (lr = 1e-4)
Loss	Categorical Cross-Entropy
Batch Size	64
Max Epochs	120
Val Split	15% of training data
Split	80% train / 20% test (stratified)

Callbacks

EarlyStopping

patience=12, restores best weights on val_loss

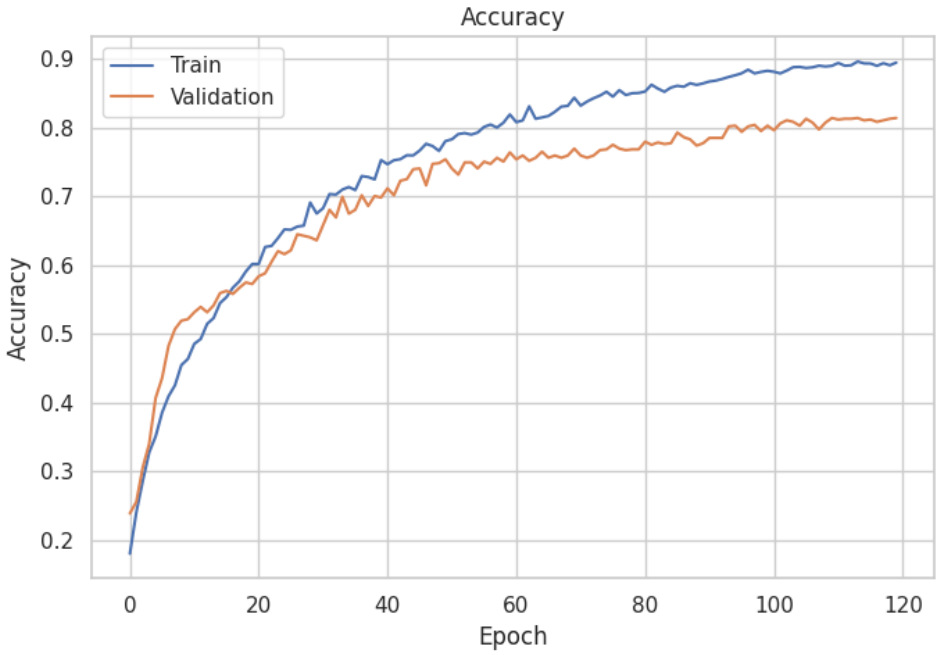
ReduceLROnPlateau

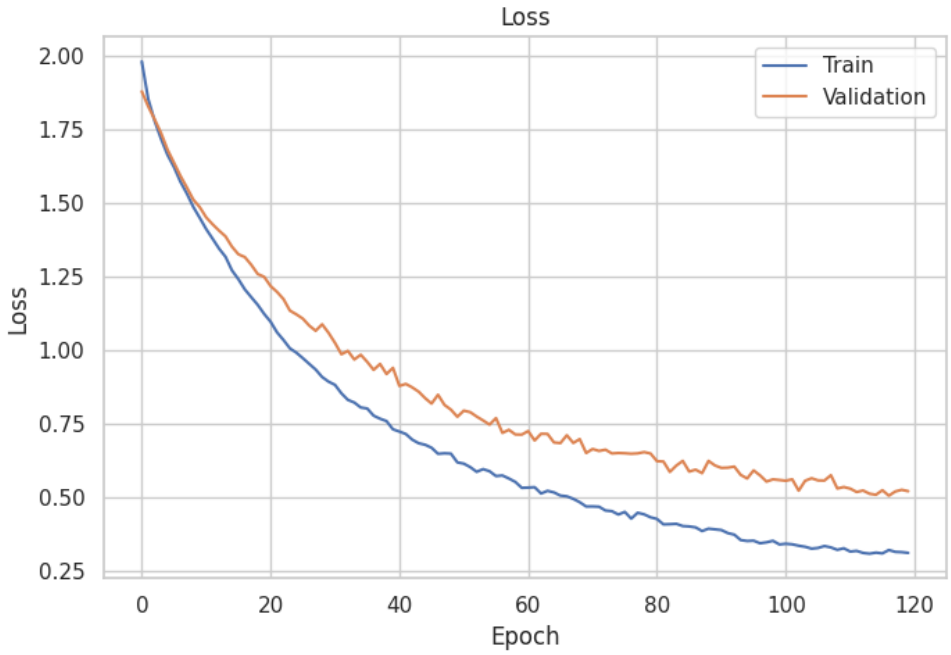
factor=0.5, patience=5, min_lr=1e-7

ModelCheckpoint

saves best model by val_accuracy

Training Results





~92%

Test Accuracy

0.91

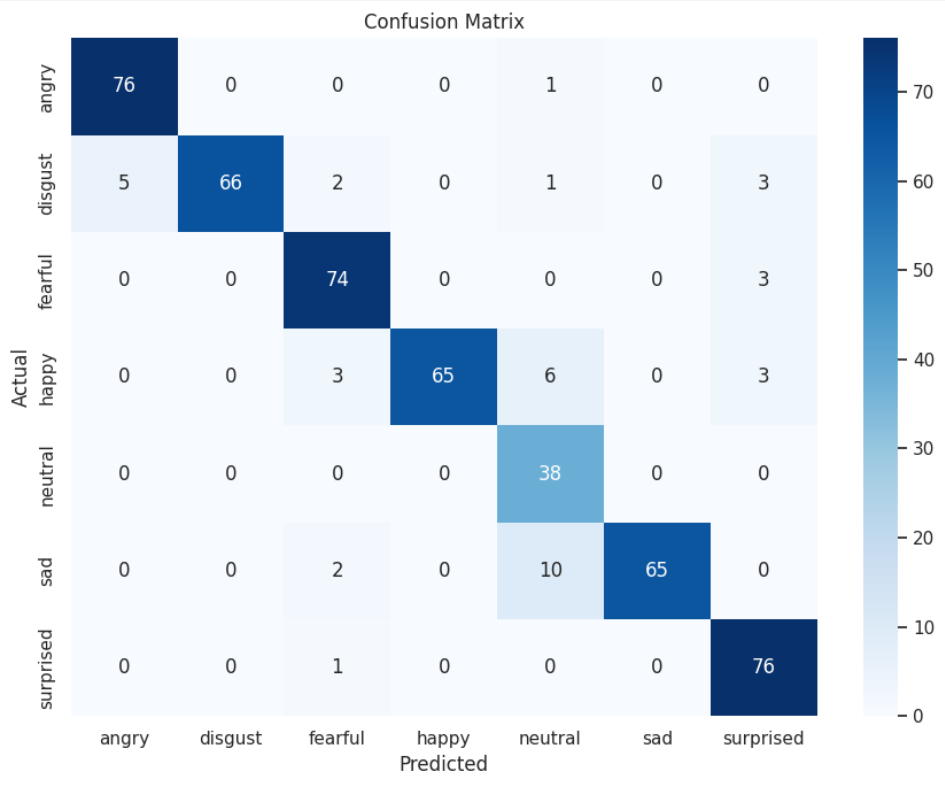
Macro F1-Score

~83%

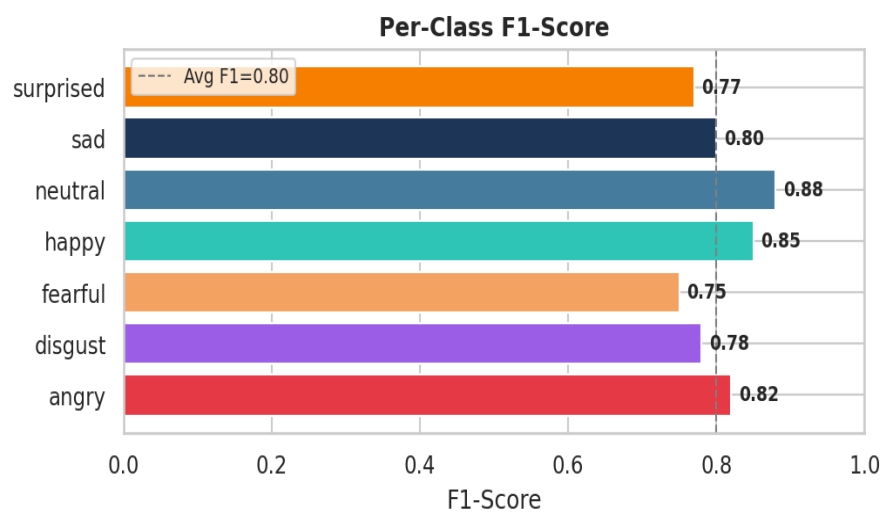
Best Val Acc

Model Evaluation & Results

Confusion Matrix



Per-Class F1-Score



Key Observations

- Neutral achieves highest F1 (0.88)
- Fearful is hardest to distinguish (0.75)
- Model generalises well with no class collapse